

In 1952, a young economics student at the University of Chicago, Harry Markowitz, developed an ingenious way to construct efficient portfolios, which he defined as those offering the highest expected return at any given level of risk (volatility), or, conversely, the lowest amount of risk at any specified level of expected return. Markowitz had taken an idea out of the realm of operations research (quadratic programming) to create an optimization algorithm that he used to map out the “frontier” of these efficient portfolios. Prior to this, there had been no quantitative way of simultaneously using expected return, volatility, and correlation to determine optimal portfolio combinations. Markowitz called his methodology mean-variance optimization (MVO).

During the oral exam defending his Ph.D. thesis, Markowitz was challenged by Milton Friedman for over an hour. Friedman argued that Markowitz’s research was not economics, business administration, or even mathematics. Markowitz received his PhD anyway and went on to become the father of modern portfolio theory. He received the Nobel Prize in Economic Sciences mainly for his PhD thesis work.

On the practical side of things, there are problems, however, in implementing MVO. As is common with many economic models, the assumptions underlying MVO do not fit the real world very well.² MVO results are unstable when the covariance (the combination of correlation and volatility) matrix is ill conditioned, which is brought about by having very similar assets. MVO results are also highly sensitive to the inputs used. These can give unreliable results, since MVO maximizes the estimation errors of these inputs. Small input differences can lead to large output differences. This has led to MVO creating error maximizing portfolios.

Users of MVO often have to adjust the inputs, constrain them to reduce sampling error, or incorporate prior information to shrink the estimates back to values that are more reasonable. Return inputs are particularly unreliable, and some MVO users ignore them entirely by opting to use instead a minimum variance portfolio. DeMiguel, Garlappi, and Uppal (2009) showed that any gains from optimal diversification were more than offset by estimation errors. MVO produces extreme weights that fluctuate substantially over time and performs poorly out of sample.³ Equal portfolio weighting is usually superior to MVO as a practical matter.⁴ While MVO theory was elegant and impressive, like many other modern financial